

Презентация

Лабораторная работа №1

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Информация

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Приобретение практических навыков установки Rocky Linux на виртуальную машину с помощью инструмента Vagrant.

Для начала создадим папку с инициалами, в которой будет 2 папки.

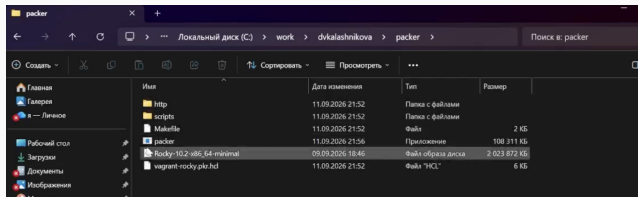


Рис. 1: Создание папок

Инициализация packer

Теперь инициализируем packer и сделаем билд образа.

```
C:\Program Files\Far Manager>cd C:\work\dkalashnikova
C:\work\dkalashnikova>packer>packer.exe init vagrant-rocky.pkr.hcl
Installed plugin github.com/hashicorp/qemu v1.1.3 in "C:/Users/dora2/AppData/Roaming/packer.d/plugins/github.com/hashicorp/qemu/packer-plugin-qemu_v1.1.3_x5_0_windows_and64.exe"
Installed plugin github.com/hashicorp/vagrant v1.1.3 in "C:/Users/dora2/AppData/Roaming/packer.d/plugins/github.com/hashicorp/vagrant/packer-plugin-vagrant_v1.1.3_x5_0_windows_and64.exe"
Installed plugin github.com/hashicorp/virtualbox v1.1.2 in "C:/Users/dora2/AppData/Roaming/packer.d/plugins/github.com/hashicorp/virtualbox/packer-plugin-virtualbox_v1.1.2_x5_0_windows_and64.exe"
C:\work\dkalashnikova>packer>packer.exe build vagrant-rocky.pkr.hcl
virtualbox-iso rockylinux: output will be in this color.
qemu.rockylinux: output will be in this color.
Build 'qemu.rockylinux' errored after 4 milliseconds 386 microseconds: Failed creating Qemu driver: exec: "qemu-system-x86_64": executable file not found in %PATH%
==> virtualbox-iso.rockylinux: Retrieving Guest additions
==> virtualbox-iso.rockylinux: Trying C:\Program Files\Oracle\VirtualBox\BoxGuestAdditions.iso
==> virtualbox-iso.rockylinux: Trying file://C:/Program%20Files/Oracle/VirtualBox/BoxGuestAdditions.iso
==> virtualbox-iso.rockylinux: file://C:/Program%20Files/Oracle/VirtualBox/BoxGuestAdditions.iso ==> C:/Program Files/Oracle/VirtualBox/BoxGuestAdditions.iso
==> virtualbox-iso.rockylinux: Retrieving ISO
==> virtualbox-iso.rockylinux: Trying Rocky-10.2-x86_64-minimal.iso
==> virtualbox-iso.rockylinux: Trying Rocky-10.2-x86_64-minimal.iso?checksumsha256%3Aa5ac6a3ce781b91a91ce78463485f66c611a5dca4b3848c779e5e81d97382f6c8
==> virtualbox-iso.rockylinux: Trying Rocky-10.2-x86_64-minimal.iso?checksumsha256%3Aa5ac6a3ce781b91a91ce78463485f66c611a5dca4b3848c779e5e81d97382f6c8 ==> C:/work/dkalashnikova/packer/rocky-10.2-x86_64-minimal.iso
==> virtualbox-iso.rockylinux: Starting HTTP server on port 8970
==> virtualbox-iso.rockylinux: Creating virtual machine...
==> virtualbox-iso.rockylinux: Creating hard drive output-rockylinux10-virtualbox/rockylinux10-virtualbox.vdi with size 61440 MiB...
==> virtualbox-iso.rockylinux: Mounting ISOs...
```

Рис. 2: Инициализация packer

Добавление образа в vagrant

После этого добавим его в vagrant.

```
C:\work\dvkashnikova\code\vagrant>vagrant box add rockylinux10 vagrant-virtualbox-rockylinux10-x86_64.box
==> box: Box file was not detected as metadata. Adding it directly...
==> box: Adding box 'rockylinux10' (v0) for provider: (and64)
box: Unpacking necessary files from: file://C:/work/dvkalashnikova/code/vagrant/vagrant-virtualbox-rockylinux10-x86_64.box
box:
==> box: Successfully added box 'rockylinux10' (v0) for '(and64)'!

C:\work\dvkashnikova\code\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Preparing master VM for linked clones...
server: This is a one time operation. Once the master VM is prepared,
server: it will be used as a base for linked clones, making the creation
server: of new VMs take milliseconds on a modern system.
```

Рис. 3: Добавление образа в vagrant

Запустим через vagrant VM сервера.

```
C:\work\dvkalashnikova\code\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Preparing master VM for linked clones...
server: This is a one time operation. Once the master VM is prepared,
server: it will be used as a base for linked clones, making the creation
server: of new VMs take milliseconds on a modern system.
==> server: Importing base box 'rockylinux10'...
```

Рис. 4: Запуск сервера

И запустим еще клиент.

```
C:\work\dvkalashnikova\code\vagrant>vagrant up client
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Cloning VM...
==> client: Matching MAC address for NAT networking...
==> client: Setting the name of the VM: vagrant_client_1789463921822_4852
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
      client: Adapter 1: nat
      client: Adapter 2: intnet
==> client: Forwarding ports...
      client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
      client: SSH address: 127.0.0.1:2200
      client: SSH username: vagrant
      client: SSH auth method: password
```

Рис. 5: Запуск клиента

Проверка через GUI

Убедимся, что они оба работают, через графический интерфейс. Войдём туда под пользователем vagrant.

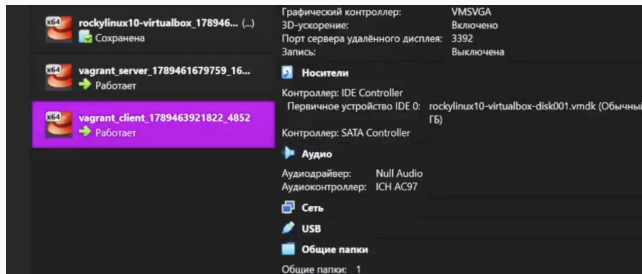


Рис. 6: Вход через GUI

Подключение по SSH

Теперь попробуем зайти на сервер через ssh, после чего авторизируемся от имени собственного пользователя, и отключимся.

```
C:\work\dvkalashnikova\code\vagrant>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
vagrant@127.0.0.1's password:
vagrant@127.0.0.1's password:
Last failed login: Tue Sep 15 10:17:18 UTC 2026 from 10.0.2.2 on ssh:notty
There were 2 failed login attempts since the last successful login.
Last login: Tue Sep 15 10:07:47 2026
vagrant@server:~$ su - dvkalashnikova
Password:
[dvkalashnikova@server.dvkalashnikova.net ~]$ exit
```

Рис. 7:>Login на сервере

Сделаем то же самое для клиента, выключим обе машины.

```
C:\work\dvkalashnikova\code\vagrant>vagrant halt server
==> server: Attempting graceful shutdown of VM...
==> server: Forcing shutdown of VM...

C:\work\dvkalashnikova\code\vagrant>vagrant halt client
==> client: Attempting graceful shutdown of VM...
```

Рис. 8: Завершение работы

Также убедимся, что у нас всё корректно в файле Vagrantfile.

```
## Common configuration
config.vm.provision "common dummy", type: "shell", preserve_order: true, path: "provision/default/01-dummy.sh"
config.vm.provision "common hostname", type: "shell", preserve_order: true, run: "always", path: "provision/default/01-hostname.sh"
```

Рис. 9: Проверка файла Vagrantfile

В результате выполнения лабораторной работы были получены навыки работы с vagrant.